

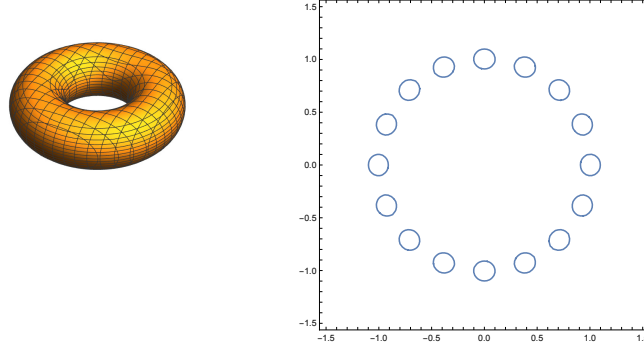


FIGURE 2. The left one is the nodal sets of U_* inside the cube $[-5/2, 5/2]^3$ when $m = 16$ and $d_m = \sqrt{2}m \log m \left(\sum_{j=1}^{m-1} \csc \frac{j\pi}{m} \right)^{-1}$. The right one is the intersection of nodal sets of q_m with $z_1 z_2$ -plane.

and

$$\zeta_j(z) = \begin{cases} \zeta(m\eta^{-1}|z|^{-2}|(z - \xi_j|z|)|), & \text{if } |z| > 1, \\ \zeta(m\eta^{-1}|z - \xi_j|), & \text{if } |z| \leq 1. \end{cases}$$

Here $\zeta(s)$ is a smooth function such that $\zeta(s) = 1$ for $s < 1$ and $\zeta(s) = 0$ for $s > 2$, and η is a small positive constant. It is not difficult to check that

$$(2.8) \quad U(z) = \frac{1}{|z|} U\left(\frac{z}{|z|^2}\right), \quad U_j(z) = \frac{1}{|z|} U_j\left(\frac{z}{|z|^2}\right), \quad j = 1, \dots, m,$$

and

$$(2.9) \quad \zeta_j(z) = \zeta_j\left(\frac{z}{|z|^2}\right), \quad j = 1, \dots, m.$$

Together with the properties that the ansatz is $2\pi/m$ -rotationally invariant in $z_1 z_2$ plane and even symmetry with respect to the z_3 axis, one can solve ψ from (2.7) uniquely in a suitable Sobolev space, which is the orthogonal complement of the space spanned by the kernels of the linear operator $\Delta + 5U^4$. More precisely, the authors in [20] proved that

$$\|\psi\|_{L_w^\infty} \leq C \left(\|V(z)\psi\|_{L_w^q} + 5 \left\| |U_*|^4 \sum_j (1 - \zeta_j) \tilde{\phi}_j \right\|_{L_w^q} + \|M(\psi)\|_{L_w^q} \right),$$

where

$$\|\psi\|_{L_w^\infty} := \|(1 + |z|)\psi\|_{L^\infty(\mathbb{R}^3)} \quad \text{and} \quad \|h\|_{L_w^q} := \|(1 + |z|)^{5-\frac{6}{q}}h\|_{L^q(\mathbb{R}^3)}.$$

Particularly, for $q > \frac{3}{2}$ it has been shown that

$$\left\| \left(1 - \sum_{j=1}^m \zeta_j \right) N \left(\sum_{j=1}^m \tilde{\phi}_j + \psi \right) \right\|_{L_w^q} + \left\| |U_*|^4 \sum_j (1 - \zeta_j) \tilde{\phi}_j \right\|_{L_w^q} + \|V(z)\psi\|_{L_w^q} \leq \frac{C}{(\log m)^2}.$$